DATA SCIENCE

PROJECT

EDA & Patient Segmentation of Diabetes Health Indicators Using K-Means Clustering

1. Introduction

Diabetes mellitus is among the most severe public health crises of the 21st century. The International Diabetes Federation estimates that 537 million adults lived with diabetes globally in 2021, projected to reach 784 million by 2045, with associated healthcare costs exceeding USD 966 billion, a 316% rise over the prior decade (Iparraguirre-Villanueva et al., 2023). Eighty-one percent of this burden falls on low- and middle-income countries, where late detection and limited infrastructure drive complications including blindness, kidney failure, stroke, and amputation (WHO, 2023). This study applies Exploratory Data Analysis (EDA) and K-Means clustering to the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset to uncover patterns in diabetes risk factors and segment the population into clinically actionable groups. The analysis is guided by the anchor article Iparraguirre-Villanueva et al. (2023), which benchmarked five supervised ML classifiers on a diabetes dataset, identifying BMI, age, blood pressure, and cholesterol as key predictors. Where that study focused on binary prediction (K-NN: 79.6% accuracy; BNB: 77.2% accuracy), the present work extends insight generation to unsupervised population segmentation using a substantially larger and more diverse dataset (n = 253,680 vs. n = 768). All analyses were conducted in R within the RStudio IDE using the CRISP-DM process model.

2. Dataset Description

Attribute	Detail
Source	https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset
File	diabetes_binary_health_indicators_BRFSS2015.csv
Observations	253,680 rows | 22 variables | After deduplication: 229,787 unique rows
Target Variable	Diabetes_binary (0 = No Diabetes, 1 = Diabetes/Pre-diabetes)
Class Split	~85.4% No Diabetes | ~14.6% Diabetes (imbalanced)
Working Sample	5,000 rows randomly drawn (set.seed = 123)
Key Variables	BMI, Age, HighBP, HighChol, GenHlth, PhysActivity, MentHlth, PhysHlth, Income, Smoker

3. Data Preprocessing

Following the CRISP-DM data preparation phase, three steps were applied. First, a missing value check via colSums(is.na()) confirmed no missing values across all 22 variables, consistent with CDC's quality assurance procedures. Second, sum(duplicated()) identified 23,893 duplicate rows (9.4% of raw data), which were removed, leaving 229,787 clean records. Third, a reproducible random sample of 5,000 observations was drawn (sample_n(), seed = 123) for EDA and clustering. The target variable is notably imbalanced, approximately 85.4% non-diabetic, which is acknowledged as a limitation and mirrors real-world population prevalence (Iparraguirre-Villanueva et al., 2023).

4. Exploratory Data Analysis

EDA was conducted across seven visualisations using ggplot2, interrogating relationships between diabetes status and BMI, age, general health, risk factors, and income, variables identified as key predictors by Iparraguirre-Villanueva et al. (2023).

4.1 BMI Distribution

Figures 1 and 2 show that non-diabetic individuals cluster between BMI 22–30 (healthy to slightly overweight), while diabetic individuals show a right-shifted distribution with greater density in the 28–38 range. The boxplot confirms the diabetic group's median BMI is approximately 5 units higher with a wider IQR. This aligns directly with Iparraguirre-Villanueva et al. (2023), who identified BMI as the strongest predictor across all five ML classifiers tested, and with Ganie & Malik (2022), who confirmed lifestyle-related obesity as a principal driver of Type 2 diabetes.

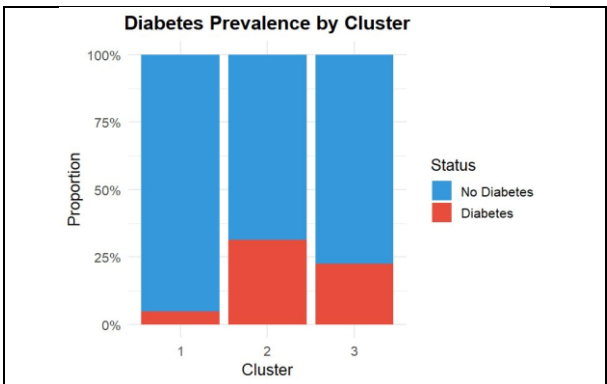

Figure 1. BMI Histogram by Diabetes Status.

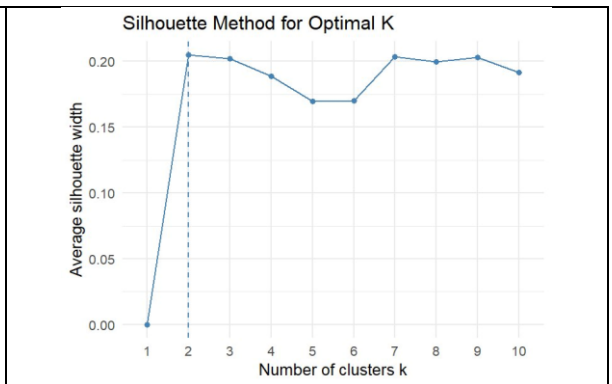

Figure 2. BMI Boxplot by Diabetes Status.

4.2 Age and General Health

Figure 3 reveals a monotonic increase in diabetes prevalence with advancing age, negligible in categories 1-3 (ages 18-34) but exceeding 20% in categories 10-13 (ages 65+). This is consistent with Theerthagiri et al. (2023), who identified age as a top ML classifier feature. Figure 4 shows a pronounced gradient by self-rated health: diabetes is near-absent among those rating health "Excellent" (category 1) but accounts for over 35% in the "Poor" category (5), a bidirectional relationship where poor health predisposes to diabetes and diabetes deteriorates health outcomes.

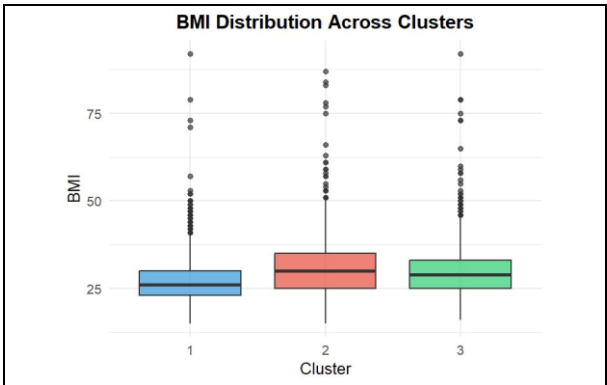

Figure 3. Diabetes Prevalence by Age Group (1=18–24; 13=80+).

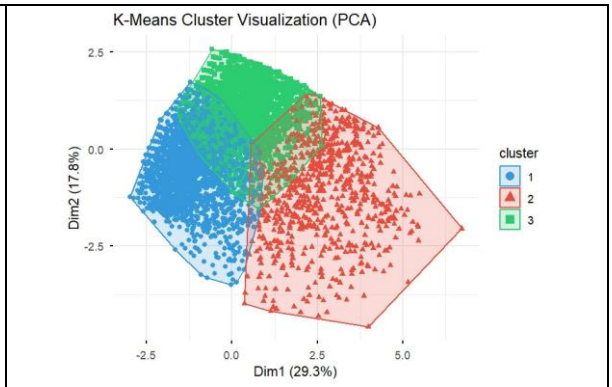

Figure 4. Diabetes Prevalence by General Health Rating (1=Excellent; 5=Poor).

4.3 Risk Factors and Income

Figure 5 compares four binary risk factors. Hypertension (HighBP) shows the largest disparity: ~79% prevalence among diabetics vs. ~41% among non-diabetics. High cholesterol (HighChol) shows a

similar gap (~70% vs. ~41%). Physical activity is inversely associated, diabetics report lower activity (~62%) than non-diabetics (~75%), confirming its protective role (Ganie & Malik, 2022). Smoking shows minimal difference (~51% vs. ~47%). Figure 6 reveals a clear negative socioeconomic gradient: diabetes prevalence in the lowest income bracket is nearly three times that of the highest, a dimension absent from the Pima Indian dataset of Iparraguirre-Villanueva et al. (2023), underscoring the value of the BRFSS dataset for equity-focused analysis (WHO, 2023).

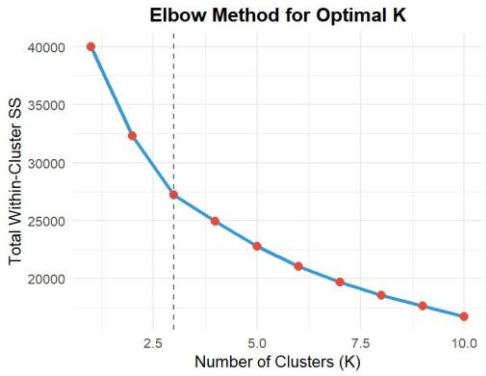

Figure 5. Risk Factor Prevalence by Diabetes Status.

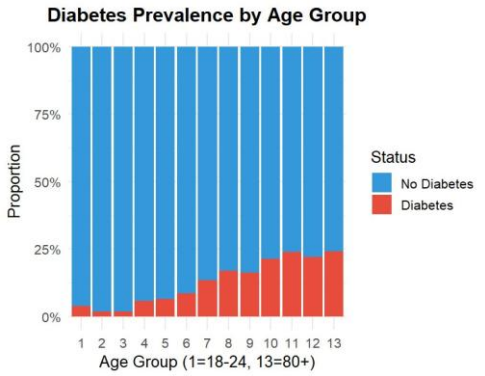

Figure 6. Diabetes Prevalence by Income Level (1=<$10k; 8=>$75k).

5. Correlation Analysis

A Pearson correlation matrix was computed for 12 variables and visualised as a heatmap (Figure 7). Diabetes_binary correlates most strongly with HighBP ($r \approx 0.30$), GenHlth ($r \approx 0.29$), Age ($r \approx 0.24$), BMI ($r \approx 0.22$), and HighChol ($r \approx 0.22$). PhysActivity shows a negative association ($r \approx -0.16$), confirming its protective role. These results closely parallel the predictor importance rankings in Iparraguirre-Villanueva et al. (2023). Notably, all individual correlations remain below |0.35|, confirming that diabetes risk is multifactorial; no single variable dominates, which justifies a multivariate clustering approach over simple bivariate analysis.

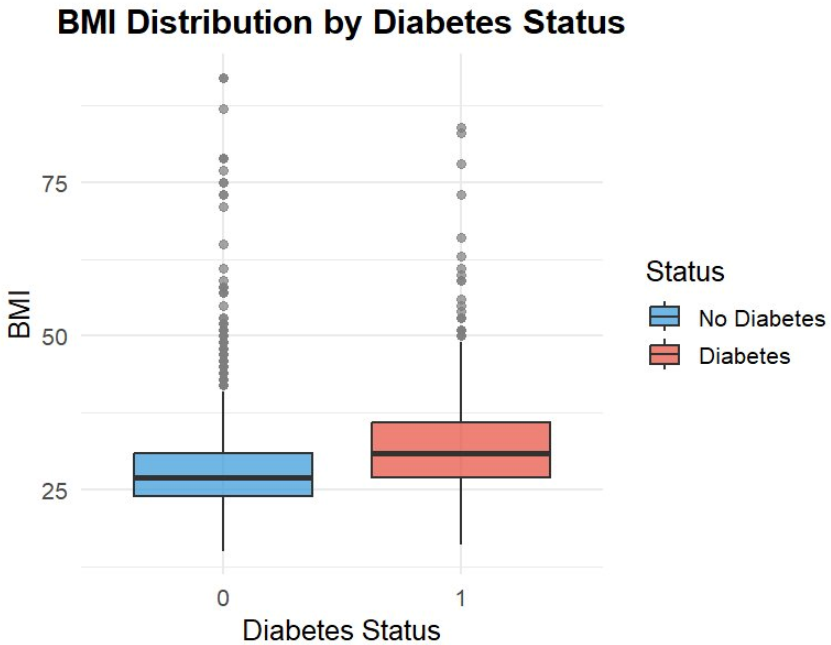

Figure 7. Correlation Heatmap of Diabetes Health Indicators (12 variables).

6. Unsupervised ML: K-Means Clustering

K-Means was selected for its computational efficiency, scalability, and wide application in healthcare population segmentation (Saxena et al., 2022). It directly extends the unsupervised strand initiated by PCA in Iparraguirre-Villanueva et al. (2023) from dimensionality reduction to full population segmentation.

6.1 Variable Selection and Scaling

Eight variables were selected based on theoretical relevance and correlation findings: BMI, Age, GenHlth, MentHlth, PhysHlth, Income, HighBP, and HighChol. All were standardised using scale() (mean = 0, SD = 1). Standardisation is critical for Euclidean distance-based algorithms to prevent high-range variables from dominating cluster assignments , the same rationale applied in the normalisation step of Iparraguirre-Villanueva et al. (2023).

6.2 Optimal K: Elbow and Silhouette Methods

Two methods were used to determine k. The Elbow Method (Figure 8) plots WCSS against k = 1–10; a clear inflection occurs at k = 3, where the reduction rate slows markedly (WCSS drops from ~32,500 at k = 2 to ~27,000 at k = 3, then diminishing returns). The Silhouette Method (Figure 9) peaks at k = 2 (≈ 0.205) with k = 3 nearly equal (≈ 0.204). Given the minimal quality difference and the substantially richer clinical interpretive value of three segments, k = 3 was selected , consistent with standard three-tier public health risk stratification practice (Jain, 2010).

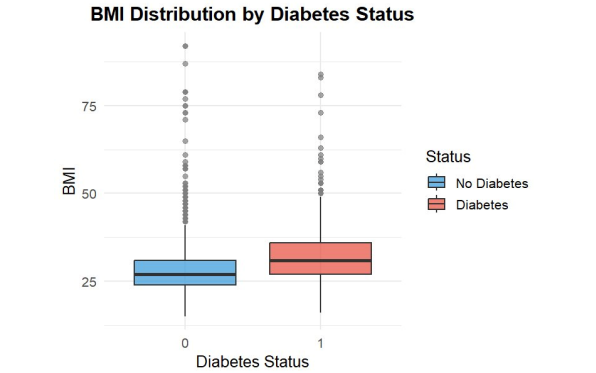

Figure 8. Elbow Method , inflection at k = 3.

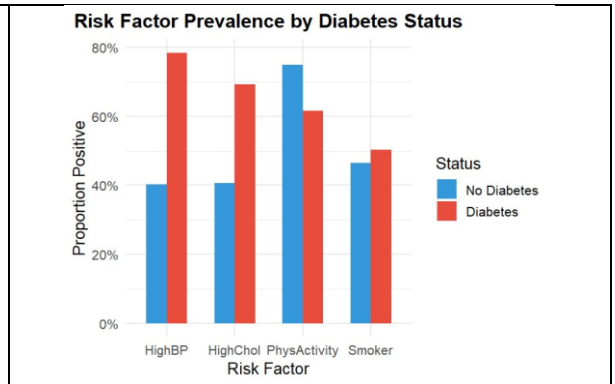

Figure 9. Silhouette Method , k = 2 and k = 3 near-equal.

6.3 Cluster Visualisation

Figure 10 shows a PCA-based 2D projection of the three clusters. Dim1 (29.3%) and Dim2 (17.8%) together explain 47.1% of variance. The clusters are visually distinct, validating the k = 3 solution.

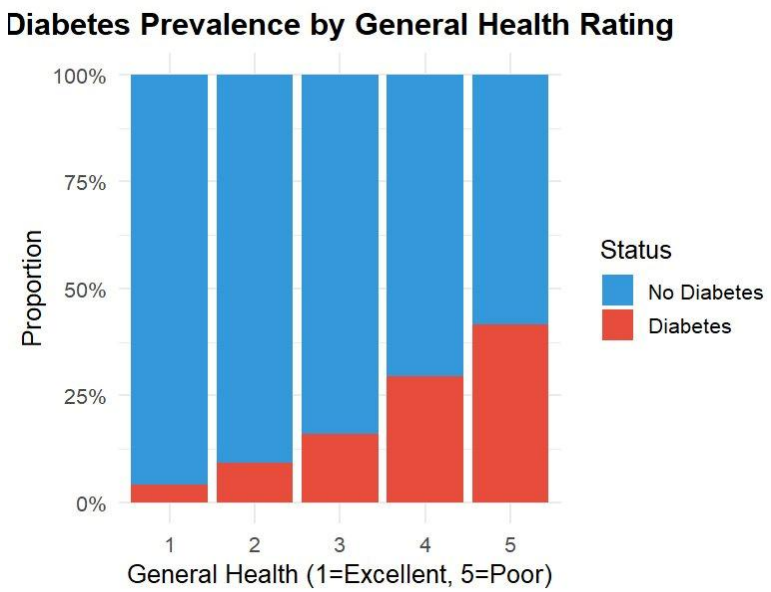

Figure 10. K-Means Cluster Visualisation (PCA). Dim1 = 29.3%, Dim2 = 17.8%.

6.4 Cluster Profiles

Characteristic	Cluster 1, Low Risk	Cluster 2, High Risk	Cluster 3, Moderate Risk
Size	~2,050 obs.	~1,550 obs.	~1,400 obs.
Avg. BMI	Lower (~26)	Higher (~31)	Moderate (~30)
Age Category	Younger (4–5)	Older (8–9)	Middle (7–8)
General Health	Better (~2.0)	Poorer (~3.5)	Moderate (~3.0)
High BP	~20%	~70%	~55%
High Cholesterol	~25%	~65%	~55%
Income Level	Higher	Lower	Moderate
Diabetes Prevalence	~7%	~32%	~22%

Table 1. K-Means Cluster Profiles (k = 3).

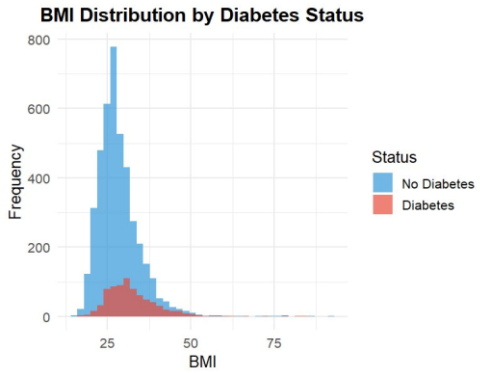

Figure 11. Diabetes Prevalence by Cluster.

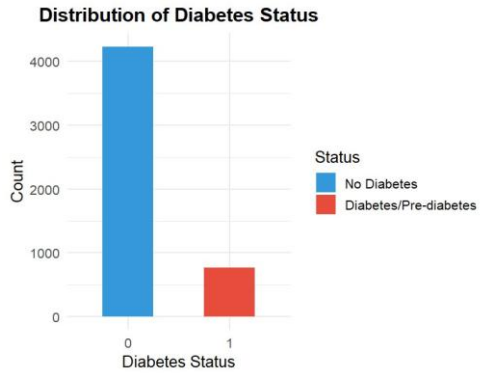

Figure 12. BMI Distribution Across Clusters.

Cluster 1 - Low-Risk Group: Youngest, lowest BMI (~26), best general health, highest income, lowest comorbidity rates (BP ~20%, Chol ~25%), diabetes prevalence ~7%. Appropriate for primary prevention and health maintenance programmes.

Cluster 2 - High-Risk Group: Highest diabetes burden (~32%). Oldest, elevated BMI (~31), severe comorbidities (BP ~70%, Chol ~65%), lowest income, poorest health. This profile mirrors the high-risk patients targeted by the K-NN and BNB classifiers of Iparraguirre-Villanueva et al. (2023). Requires immediate clinical intervention and intensive disease management.

Cluster 3 - Moderate-Risk Group: Intermediate risk (~22% diabetes), middle-aged, moderate BMI and comorbidities. Most responsive to preventive lifestyle interventions (structured physical activity, dietary modification) that could arrest progression toward the Cluster 2 profile.

7. Discussion

The clustering results converge with and extend Iparraguirre-Villanueva et al. (2023) across two key dimensions. First, the variables most discriminating in the K-Means solution, BMI, age, blood pressure, cholesterol, and general health, mirror the predictor importance rankings across all five supervised ML classifiers in the anchor article. This cross-methodological, cross-dataset convergence strengthens confidence in these variables as robust diabetes risk markers. Second, the income gradient documented in the EDA and reflected in Cluster 2's lower income profile introduces a socioeconomic dimension absent from the Pima Indian dataset. Economic deprivation compounds clinical risk through reduced access to healthcare, nutritious food, and physical activity, a structural vulnerability that clinical prediction models alone cannot capture (WHO, 2023). A key limitation is the class imbalance (85.4% non-diabetic), which skews clustering toward non-diabetic profiles. Future work should apply SMOTE, as used by Iparraguirre-Villanueva et al. (2023), before clustering to produce more balanced segments. The moderate silhouette score (≈ 0.20) is expected in large behavioural survey data and reflects within-cluster heterogeneity inherent to population-level analyses rather than a weakness of the solution itself.

8. Conclusion

This study applied EDA and K-Means clustering (k = 3) to the CDC BRFSS 2015 Diabetes Health Indicators dataset using the CRISP-DM framework in RStudio. Three clinically meaningful patient segments were identified: a low-risk group (7% diabetes prevalence), a moderate-risk group (22%), and a high-risk group (32%). Key risk discriminators, BMI, age, blood pressure, cholesterol, general health, and income, align closely with the predictor findings of Iparraguirre-Villanueva et al. (2023), providing cross-methodological validation. The segmentation offers direct business value for healthcare planners: Cluster 2 requires immediate clinical investment; Cluster 3 offers the highest return on preventive intervention; Cluster 1 benefits from awareness and maintenance programmes.

References

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